Lecture Notes: From Inertial Dynamics to the Circular Restricted Three-Body Problem

Orbital Mechanics Course Materials

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1 The Inertial Foundation (Chapter 1)

To understand the complexity of the Three-Body Problem, we must first recall the absolute dynamics of the Two-Body Problem (2BP). In an inertial reference frame (X, Y, Z), the motion of a mass m_i is governed by Newton's Second Law and the Law of Universal Gravitation:

$$m_i \ddot{\vec{R}}_i = \sum_{j=1, j \neq i}^{n} \frac{G m_i m_j}{r_{ij}^3} \vec{r}_{ij} \tag{1}$$

For the 2BP, this leads to the conservation of specific mechanical energy ε and specific angular momentum $\vec{h}$. These quantities are invariant because the gravitational field is static (time-independent) in the inertial frame.

2 Transition to the CR3BP (Chapter 2)

The Circular Restricted Three-Body Problem (CR3BP) introduces two major simplifications to the general Three-Body Problem:

1. **Restricted:** We assume the third body (m_3, the spacecraft) has negligible mass ($m_3 \to 0$). Thus, it does not influence the orbits of the two primaries (m_1 and m_2).
2. **Circular:** We assume the primaries revolve around their common center of mass (barycenter) in perfect circular orbits.

2.1 The Rotating (Synodic) Frame

Because the gravitational field of m_1 and m_2 is rotating in the inertial frame, the energy of m_3 is not conserved in that frame. To find a constant of motion, we transform our equations into a **Synodic Frame** (x, y, z) that rotates with an angular velocity $\vec{\omega}$ equal to the orbital mean motion of the primaries.

Using the kinematic relations for a rotating frame (*Curtis Section 1.7*):

$$\vec{a}_{inertial} = \vec{a}_{rel} + 2\vec{\omega} \times \vec{v}_{rel} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \tag{2}$$

Here $\vec{v}_{rel}$ and $\vec{a}_{rel}$ are the velocity and acceleration of the body with respect to the rotating coordinate frame. In other words, this is the PDE that governs the motion in the rotating frame.

$$\ddot{\vec{r}}_{inertial} = \ddot{\vec{r}}_{rel} + 2\vec{\omega} \times \dot{\vec{r}}_{rel} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \tag{3}$$

In the CR3BP, this results in the appearance of two "fictitious" accelerations: the **Coriolis acceleration** $(2\vec{\omega} \times \vec{v}_{rel})$ and the **Centrifugal acceleration** $(\vec{\omega} \times (\vec{\omega} \times \vec{r}))$.

3 The Jacobi Constant: Conservation in Chaos

In the rotating frame, the equations of motion are time-independent. By taking the dot product of the equations of motion with the velocity vector $\vec{v} = [\dot{x}, \dot{y}, \dot{z}]^T$ and integrating with respect to time, we derive the **Jacobi Constant** (C).

The expression for C is often written in terms of the **Effective Potential** (or Pseudo-Potential), Ω:

$$C = 2\Omega(x, y, z) - v^2 \tag{4}$$

where:

$$\Omega(x, y, z) = \underbrace{\frac{1}{2}\omega^2(x^2 + y^2)}_{\text{Centrifugal Potential}} + \underbrace{\frac{\mu_1}{r_1} + \frac{\mu_2}{r_2}}_{\text{Gravitational Potential}} \tag{5}$$

Note: Unlike the 2BP, where energy is a property of the orbit, the Jacobi Constant is the *only* conserved scalar in the CR3BP. It defines the **Zero Velocity Surfaces** (where $v = 0$) that dictate where a spacecraft can and cannot travel.

4 Glossary of Terms

Barycenter The center of mass of the two primary bodies. In the CR3BP, the origin of the synodic frame is placed here.

Libration Points Another name for Lagrange points. These are equilibrium points where the gravitational and centrifugal forces perfectly cancel out in the rotating frame.

Mass Parameter (π_2) The ratio of the secondary mass to the total system mass: $\pi_2 = m_2/(m_1 + m_2)$. This determines the geometry of the Lagrange points.

Pseudo-Potential (Ω) A scalar field representing the combined effects of gravity and the centrifugal force in a rotating frame.

Synodic Frame A non-inertial reference frame that rotates at the same rate as the binary system, making the two primaries appear stationary.

Zero Velocity Curves (ZVC) The boundary lines in the x-y plane where a spacecraft with a specific Jacobi Constant would have zero velocity. These act as "forbidden zone" boundaries.

5 Practical Problem: Visualizing the Gateways

Refer to the *CR3BP Interactive Simulator*.

1. Observe the "Saddle" points at L_1 and L_2. These are the lowest energy points where the ZVCs first "break open" to allow passage.

2. For a spacecraft to move from the Primary to the Secondary body, its Jacobi Constant must be **lower** than the potential at L_1 ($C < C_{L1}$).